

PROFESSIONAL SUMMARY

Software Engineer with 7+ years of experience building scalable, distributed systems across Amazon Ads and enterprise platforms. Currently an SDE 2 at Amazon, leading end-to-end development of full-stack features spanning backend services, APIs, frontend systems, and cloud infrastructure.

Proven track record of delivering complex systems including access control platforms, batch APIs, observability pipelines, and advertiser-facing products at scale. Strong expertise in system design, microservices, and cloud-native architectures (AWS/CDK).

Additionally experienced in AI/ML systems, including LLMs, RAG pipelines, and intelligent automation, with a Master's degree in Machine Learning from Carnegie Mellon.

Focused on building high-impact, reliable systems and driving engineering excellence across the stack.

EDUCATION

- Carnegie Mellon University - School of Computer Science** Pittsburgh, PA
Master of Science in Machine Learning, School of Computer Science; GPA: 3.4/4.0 Aug. 2019 – Dec. 2020
- Indian Institute of Technology, Kharagpur** West Bengal, India
Bachelor of Engineering in Computer Science and Engineering; GPA: 4.0 (9.72/10.0) Aug. 2013 – July. 2017

PROFESSIONAL EXPERIENCE

- Amazon** Seattle, WA
SDE 2 – Amazon Ads Mar 2025 – Present
 - Full-Stack Ownership of Ads Permission Delegation Platform:** Designed and delivered multiple end-to-end features spanning API models, backend services, website backend, React frontend, and CDK infrastructure across ~18 packages. Owned access level management, business justification workflows, and source validation across advertiser account management systems — leading design, implementation, and code review across the full stack.
 - Batch APIs with Concurrent Execution & Fault Tolerance:** Architected and implemented batch ADD/DELETE APIs supporting parallel multi-threaded execution of mapping operations. Designed thread-safe processing with fine-grained concurrency control, partial success/failure isolation per mapping, and a tiered error classification system. Implemented exponential backoff with jitter for transient failures, idempotency guarantees for retries, and CloudWatch-backed status metrics for real-time batch health monitoring.
 - ML Inference Integration & KPI Widget Platform:** Built a micro-frontend KPI visualization widget from scratch using React and Webpack Module Federation with CDN delivery, i18n support, and CDK-based deployment pipelines. Extended the widget to support dynamic metric visualization by integrating real-time ML model inference — consuming inference endpoints from a partner ML platform team — enabling live, model-driven KPI computation with sparklines and responsive layouts for Ads Console.
 - Observability, Reliability & Security Improvements:** Built dependency service monitoring across accessor layers with CloudWatch dashboards, alarms, and structured logging, reducing mean time to detection for production incidents. Resolved Sev-2 outages via targeted validation fixes, implemented username blocklisting to address abuse vectors, and identified and mitigated dependency confusion vulnerabilities in production build pipelines.
 - Policy-Based Access Control & Identity Infrastructure:** Refactored policy configuration layer to enable always-on policy-based access control with marketplace-aware access levels and UUID-based identity mapping. Added region validation and detailed error surfacing to improve correctness and debuggability across multi-region advertiser account systems.
 - Operational Excellence & Technical Leadership:** Led system migrations including weblab deprecations, AMC config transitions, and feature flag removals. Authored 75+ production code reviews and reviewed 140+ peer changes, influencing API contracts, design patterns, and reliability practices across the Ads Permission Delegation ecosystem.
- AI Startups — Founding & Technical Leadership Roles** Remote / Multiple
 - Senior ML & founding technical leadership across three early-stage AI startups* Sep 2023 – Mar 2025
 - Zaytun AI (formerly LIVER)** San Jose, CA
Senior ML Engineer (Contract) Dec 2024 – Mar 2025
 - AI-Powered Recommendation Engine:** Designed and built a vector search-based ML recommendation system leveraging LLM embeddings to match users with personalized short-form food video content and local restaurants, supporting a TikTok-style platform targeting a global Muslim audience. Owned the full ML pipeline: embedding generation, vector indexing, retrieval ranking, and relevance tuning. Migrated from Vertex AI Vector Search to Qdrant, reducing infrastructure costs and improving deployment flexibility.
 - Scalable ML Backend for iOS Launch:** Architected and deployed backend services on Firebase and Google Cloud to support real-time ML inference delivery to mobile clients, including authentication, real-time data sync, and push notification pipelines enabling scalable content delivery and user engagement.
 - HyperWater.ai** Dallas, TX
Senior ML Engineer & Tech Lead Jan 2024 – Nov 2024

- **Generative AI & LLM Integration:** Architected and implemented an advanced ITB Preferences module leveraging LLMs and rule engines to automate decision-making processes for subcontractors. Designed the end-to-end ML pipeline from user preference ingestion to structured decision output. [\[Languages : Python, Java\]](#)
- **Agent-Based Document Processing:** Built agentic systems to automate ITB document processing — applying OCR for extraction, LLM-generated decision flows for classification and routing, and integrated APIs for ITB ingestion and action-taking workflows. Owned architecture, model integration, and production deployment.
- **LLM-Driven Workflow Optimization:** Engineered a scalable system integrating LLMs with Drools rule engine to convert unstructured user preferences into structured decision flows, improving throughput and consistency of document processing pipelines.

CeraMD.ai

Chief Technical Officer

Boston, MA

Sep, 2023 - Dec, 2023

- **Large-Scale AI System Architecture:** Designed and implemented a comprehensive AI-driven medical care system, integrating chat services, web crawlers, and advanced NLP models.
- **Generative AI and RAG Implementation:** Spearheaded the development of a retriever-based system utilizing OpenAI and LLaMA models, enabling real-time, evidence-based medical insights. [\[Languages : Python, Node.js\]](#)
- **Cloud Infrastructure and DevOps:** Orchestrated a robust cloud architecture on AWS, implementing microservices, serverless functions, and efficient data pipelines for scalable AI operations. [Website](#), [Web-app](#)

Microsoft USA

Software Engineer 2

Bellevue, WA

May, 2021 - July, 2023

- **Office Excel Charting Space:** Spearheaded telemetry and dashboard development, revolutionizing data visualization and analysis processes. Architected comprehensive telemetry dashboards and optimized data querying mechanisms, significantly enhancing product monitoring capabilities. [\[Languages : C++, C#\]](#)
- **Azure Software Defined Networking:** Led implementation of Network Security Groups on Private Link Connections using SDN Appliance Middle box, enhancing network security and performance. [\[Languages : C++, C#\]](#)

Qeexo

ML Research Engineer Intern

Pittsburgh, USA

June 2020 – Aug 2020

- **AutoML for Embedded Sensor Data:** Designed and prototyped real-time clustering and visualization pipelines for an AutoML platform targeting on-device and off-device execution on embedded hardware (Arduino Sensorboard). Built ML components for sensor data ingestion, unsupervised clustering, and result visualization, enabling non-expert users to train and evaluate models on resource-constrained devices.

Adobe Systems

Senior Software Development Engineer (SDE II)

Noida, India

June, 2017 - Aug, 2019

- **Vector Graphics Editing (iPad):** Developed core components of a vector graphics editing application for iPad, implementing advanced tools including Pen, Direct Selection, and Snapping, enabling precise design workflows and smooth transition of desktop-grade capabilities to mobile platforms.
- **Adobe Experience Manager (AEM) Platform:** Architected and built full-stack solutions for Interactive Communication applications, designing data integration services and document fragment-based systems to support scalable, cloud-based business process automation.

Adobe Big-Data Experience Lab (BEL), Adobe Research

Research Intern

Bangalore, India

May 2016 - July, 2016

- **Channel-specific Propensity Modelling:** Developed an innovative machine learning solution for multi-channel customer behavior prediction, implementing Canonical Correlation Analysis (CCA) to enhance Adobe Analytics capabilities.

PROJECTS

- **Generative AI for Music Creation (ML Term Project, CMU):** Developed advanced deep learning models using GANs and VAE-GANs for automated music generation, demonstrating proficiency in cutting-edge AI techniques.
- **Large-Scale Text Analysis with Apache Spark (Big Data Project, CMU):** Engineered a distributed computing solution for processing and analyzing the complete works of William Shakespeare, showcasing expertise in big data technologies and scalable AI systems.
- **Entity Resolution in Large Datasets (Big Data Project, CMU):** Designed and implemented a scalable entity resolution system using Spark, applying advanced analytics techniques such as TF-IDF and cosine similarity metrics to optimize text similarity calculations across multiple data sources.
- **Machine Learning Pipeline for Song Year Prediction (Big Data Project, CMU):** Constructed an end-to-end machine learning pipeline to predict song release years based on audio features from the Million Song Dataset, optimizing linear regression models using gradient descent and grid search techniques.
- **Researcher Recommendation System (IR Term Project, IIT Kharagpur):** Developed an AI-driven search and recommendation engine for the Scientific Research Community, implementing advanced clustering algorithms to suggest potential collaborators based on co-author graph analysis.
- **Citation Analysis using Deep Neural Networks (Bachelors Thesis Project, IIT Kharagpur):** Engineered a deep learning framework to analyze and classify citation sentiment in academic papers, demonstrating proficiency in natural language processing and sentiment analysis techniques.

PUBLICATIONS

- **Towards Frequency-Based Explanation for Robust CNN:** Led research on analyzing the connection between frequency components in input data and CNN reasoning processes, providing quantitative analysis of frequency contributions to model predictions. [[Publication URL](#)]
- **Supporting Throughput Fairness in IEEE 802.11ac Dynamic Bandwidth Channel Access:** Pioneered the development of Hybrid Adaptive DBCA (HA-DBCA), an innovative mechanism for enhancing fair channel access in Dynamic Bandwidth Channel Access (DBCA) environments. [[Publication URL](#)]

KEY SKILLS

- **AI/ML:** Generative AI, LLMs, RAG systems, NLP, Computer Vision, TensorFlow, PyTorch, Scikit-learn
- **Cloud & Infrastructure:** AWS (CDK, Lambda, S3, CloudWatch), Azure, Docker, Microservices architecture
- **Programming:** Python, C++, Java, JavaScript, SQL
- **Tools & Practices:** Git, CI/CD, Agile methodologies, Data pipelines, RESTful APIs

RELEVANT COURSEWORK

- **Masters:** Machine Learning, Convex Optimization, Probabilistic Graphical Models, Deep Learning
- **Bachelors:** Information Retrieval, Natural Language Processing, Artificial Intelligence, Big Data Analytics